

AR Smart Glasses for Home Applications

Artificial intelligence (AI) has become the pinnacle of modern technology, allowing humans to operate efficiently and complete tasks more quickly. However, AI has limited daily usage in basic living environments, as AI is usually connected to a mobile phone or home assistant device. The limitation of artificial intelligence usage in living environments reduces the impact AI can have on day-to-day interactions, especially for those who are negatively impacted by disabilities.

Individuals with disabilities also face a daily challenge in home spaces. For example, the blind need assistance with navigation, object recognition, and spatial awareness. The deaf are unable to hear sounds from other people and auditory cues like doorbells, a microwave going off, or a fire alarm. These limitations overall reduce safety, increase the need for outside assistance, and make living extremely difficult.

This engineered solution, *Waveguide G502 Augmented Reality Smart Glasses with AI for Efficient Home Interactions*, will assist those who have disabilities through artificial intelligence, a waveguide AR display, and automated speakers that will enhance the ability for disabled and non-disabled individuals. The glasses use a heads-up display (HUD) that provides an easy way to connect to the AI model. This model takes in information from camera data and environmental audio, which will then convert the data into readable data that can be displayed through the waveguide and speakers. By implementing AI to detect objects, languages, and sounds while using a waveguide AR display module, these smart glasses will be a continuous mobile assistant that can help anyone with basic tasks to life-changing living conditions.